Final Information

December 13th, 2024, 7:45am – 9:25am (2hrs)

Location: Psychology 121

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Important terms

Vector field…	*inputs a point (x, y) and outputs vectors*
Conservative…	*used to describe a vector field that can be described as the gradient of a function*
Line/path integral…	*area of a function along a curve*
Path independent…	*no matter what curve you take, you will result in the same value*
Open...	*all points have a circle of points around it which are still in the region*
Connected…	*any two paths can be connected by a path in the region*
Orientable…	*a surface that has a unit normal vector at each point*

Standard assumptions

In $\mathbb{R}^2$…

$$\vec{F}(x, y) = \langle P(x, y), Q(x, y)\rangle$$

$$\vec{r}(t) = \langle x(t), y(t)\rangle$$

In $\mathbb{R}^3$…

$$\vec{F}(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z)\rangle$$

$$\vec{r}(t) = \langle x(t), y(t), z(t)\rangle$$

Change of differentials…

$$ds = |\vec{r}'(t)|dt$$

$dq = q'dt$ where q is some variable

Del operator…

$$\nabla = \langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\rangle$$

$$curl\vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

$$div\vec{F} = \nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

Formal definitions

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Path independence

The integral $\int_C \vec{F} \cdot d\vec{r} = \int_C \vec{F} \cdot \vec{r}'\, dt$ is *path independent* if, for any two paths C_1 and C_2 from point A to B,

$$\int_{C_1} \vec{F} \cdot \vec{r}_1'(t)\, dt = \int_{C_2} \vec{F} \cdot \vec{r}_2'(t) \tag{1}$$

Fundamental Theorem of Line Integrals

Let C be a smooth curve from point A to B, parametized by $\vec{r}(t)$ with $a \leq t \leq b$ such that $\vec{r}(a) = A$ and $\vec{r}(b) = B$. Suppose $\vec{F}$ is a conservative vector field with potential function f so $\vec{F} = \nabla f$ where f is differentiable and $\vec{F}$ is continuous. Then,

$$\int_C \vec{F} \cdot d\vec{r} = \int_c \nabla f \cdot d\vec{r} = f\big(\vec{r}(b)\big) - f\big(\vec{r}(a)\big) = f(B) - f(A) \tag{2}$$

Implications of the Fundamental Theorem of Line Integrals

For conservative vector fields, the value of $\int_C \vec{F} \cdot d\vec{r}$ depends only on the endpoints, not the path; $\vec{F}$ is path independent.

Consequences of the Fundamental Theorem of Line Integrals

Suppose $\vec{F}$ is conservative and C is a closed curve. Then,

$$\int_C \vec{F} \cdot d\vec{r} = f(A) - f(A) = 0 \tag{3}$$

Green's Theorem (iteration #1)

$$\int_C P\, dx + Q\, dy = \iint_D \frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} dA \tag{4}$$

Conservative nature of vector fields

If $\vec{F}$ is defined on all of $\mathbb{R}^3$ and $curl\vec{F} = 0$, then $\vec{F}$ is conservative.

Important expressions

Conservative property of vector field $\vec{F}$ given by potential function f (basic, gradient)…

$$\vec{F} = \nabla f$$

Integrating scalar functions along a curve…

$$\int_C f(x,y)\,ds = \int_{t=a}^{b} f(\vec{r}(t))|\vec{r}'(t)|\,dt$$

Finding arc-length using line integrals…

$$arclength = \int_C 1\,ds$$

Breaking line integrals into pieces…

$$\int_C f\,ds = \int_{C_1} f\,ds + \int_{C_2} f\,ds + \cdots + \int_{C_n} f\,ds$$

Integrating vector functions along a curve…

$$\int_C \vec{F}\cdot ds = \int_C \vec{F}\cdot\vec{T}\,ds = \int_{t=a}^{b} \left[\vec{F}\big(\vec{r}(t)\big)\cdot\vec{r}'(t)\right] dt$$

Alternate notations for line integrals of vector functions…

$$\mathbb{R}^2\colon \int_C \vec{F}\cdot ds = \int_C P\,dx + Q\,dy$$

$$\mathbb{R}^3\colon \int_C \vec{F}\cdot ds = \int_C P\,dx + Q\,dy + R\,dz$$

Path independence…

$$\int_{C_1} \vec{F}\cdot\vec{r}_1{}'(t)\,dt = \int_{C_2} \vec{F}\cdot\vec{r}_2{}'(t)\,dt$$

Fundamental theorem of line integrals…

$$\int_C \vec{F}\cdot d\vec{r} = \int_C \nabla f\cdot d\vec{r} = f\big(\vec{r}(b)\big) - f\big(\vec{r}(a)\big) = f(B) - f(A)$$

Conservative property of vector field $\vec{F}$ given $\vec{F} = \nabla f \ldots$

$$\mathbb{R}^2: \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

$$\mathbb{R}^3: \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \, and \, \frac{\partial P}{\partial z} = \frac{\partial R}{\partial x} \, and \, \frac{\partial Q}{\partial z} = \frac{\partial Q}{\partial y}$$

Generally, $curl\vec{F} = 0$

Green's theorem…

$$\int_C P\,dx + Q\,dy = \iint_D \frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} dA = \iint_D curl\vec{F} \cdot \hat{k}\,dA$$

Surface area…

$$S = \iint_D \|\vec{r}_u \times \vec{r}_v\|\,dudv$$

Surface integral of a scalar function…

$$\iint_S f(x,y,z)\,dS = \iint_D f(\vec{r}(u,v))\|\vec{r}_u \times \vec{r}_v\|\,dudv$$

Surface integral of a vector function; flux…

$$\iint_S \vec{F} \cdot \vec{n}\,dS = \iint_D \vec{F}(\vec{r}) \cdot (\vec{r}_u \times \vec{r}_v)\,dudv$$

Stoke's theorem…

$$\int_C \vec{F} \cdot d\vec{r} = \iint_D curl\vec{F} \cdot \vec{n}\,dA = \iint_D curl\vec{F} \cdot (\vec{r}_u \times \vec{r}_v)\,dudv$$

Divergence theorem…

$$\iint_S \vec{F} \cdot \vec{n}\,dS = \iiint_E div\vec{F}\,dV$$

Implications of curl and divergence

The state in which a vector field is incompressible…

$$div\vec{F} = 0$$

Assume $\vec{F}$ is conservative such that $\vec{F} = \nabla f$…

$$curl\vec{F} = \nabla \times \nabla f = \vec{0}$$

Application of the Laplacian operator…

$$div\vec{F} = \nabla \cdot \nabla f = \nabla^2 f = f_{xx} + f_{yy} + f_{zz}$$

Conservative nature of a vector field…

$$curl\vec{F} = 0$$

The divergence of the curl vector for a vector field…

$$div\left(curl\vec{F}\right) = 0$$

Application to Green's theorem…

$$\int_C \vec{F} \cdot d\vec{r} = \iint_D \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} dA = \iint_D curl\vec{F} \cdot \hat{k}\, dA$$

The big three

Green's theorem…

$$\int_C P\, dx + Q\, dy = \iint_D \frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} dA$$

Stoke's theorem…

$$\int_C \vec{F} \cdot d\vec{r} = \iint_D curl\vec{F} \cdot \vec{n}\, dA = \iint_D curl\vec{F} \cdot (\vec{r}_u \times \vec{r}_v)\, dudv$$

Divergence theorem…

$$\iint_S \vec{F} \cdot \vec{n}\, dS = \iiint_E div\vec{F}\, dV$$

Vector basics

Distance d between two points P_1 and P_2…

$$d\big((x_1, y_1, z_1), (x_2, y_2, z_2)\big) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Equation of a sphere with radius r and center (h, k, l)…

$$r^2 = (x - h)^2 + (y - k)^2 + (z - l)^2$$

Dot product by matrix…

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Dot product by geometry…

$$\vec{a} \cdot \vec{b} = \|\vec{a}\| \cos\theta \, \|\vec{b}\|$$

Test for orthogonality…

$$\vec{a} \cdot \vec{b} = 0 \Rightarrow \vec{a} \perp \vec{b}$$

Scalar projection…

$$comp_{\vec{a}}\vec{b} = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\|}$$

Vector projection…

$$proj_{\vec{a}}\vec{b} = \left(comp_{\vec{a}}\vec{b}\right)\left(\frac{\vec{a}}{\|\vec{a}\|}\right) = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\|^2}\vec{a}$$

Cross product by matrix…

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \hat{\imath}\begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \hat{\jmath}\begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \hat{k}\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$$

Cross product by geometry…

$$\vec{a} \times \vec{b} = \|\vec{a}\| \sin\theta \, \|\vec{b}\|$$

Test for parallel…

$$\vec{a} \times \vec{b} = 0$$

Area of a parallelogram with sides $\vec{a}$ and $\vec{b}$…

$$A = \|\vec{a} \times \vec{b}\|$$

Lines...

$$\vec{r} = \vec{r}_0 + t\vec{v}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + t \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} x_0 + ta \\ y_0 + tb \\ z_0 + tc \end{bmatrix}$$

"line = point + t*direction"

Planes with normal vector $\vec{n} = \langle a, b, c \rangle$ …

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

Limits…

$$\lim_{t \to a} \vec{r}(t) = \begin{bmatrix} \lim_{t \to a} f(t) \\ \lim_{t \to a} g(t) \\ \lim_{t \to a} h(t) \end{bmatrix}$$

Derivatives..

$$\vec{r}'(t) = \begin{bmatrix} f'(t) \\ h'(t) \\ g'(t) \end{bmatrix}$$

Indefinite integral…

$$\int \vec{r}(t)dt = \begin{bmatrix} F(t) \\ G(t) \\ H(t) \end{bmatrix} + \vec{c}$$

Definite integral…

$$\int_{t_i}^{t_f} \vec{r}(t)dt = \begin{bmatrix} \int_{t_i}^{t_f} f(t)dt \\ \int_{t_i}^{t_f} g(t)dt \\ \int_{t_i}^{t_f} h(t)dt \end{bmatrix}$$

Unit tangent vector at $t = t_0$…

$$\frac{\vec{r}'(t_0)}{\|\vec{r}'(t_0)\|} = \frac{\vec{v}(t_0)}{\|\vec{v}(t_0)\|}$$

Definite arclength…

$$L = \int_{t_i}^{t_f} \|\vec{r}'(t)\|\, dt = \int_{t_i}^{t_f} \sqrt{(f'(t)^2) + \left(g'(t)\right)^2 + \left(h'(t)\right)^2}\, dt$$

Arclength function…

$$s(t) = \int_a^t s(u)\, dt$$

Unit tangent vector…

$$\vec{T}(t) = \frac{\vec{r}'(t)}{\|\vec{r}'(t)\|}$$

Unit normal vector…

$$\vec{N}(t) = \frac{\vec{T}'(t)}{\left\|\vec{T}'(t)\right\|}$$

Unit binormal vector…

$$\vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$$

Curvature…

$$\kappa = \left\|\frac{d\vec{T}}{ds}\right\| = \frac{\left\|\vec{T}'(t)\right\|}{\|\vec{r}'(t)\|} = \frac{\|\vec{r}'(t) \times \vec{r}''(t)\|}{\|\vec{r}'(t)\|^3}$$

Partial derivatives & applications

Gradient vector of function $f(x,y,z)$…

$$\nabla f(x,y,z) = \langle f_x, f_y, f_z \rangle$$

Directional derivative of $f(x,y)$ in direction of $\vec{v} = \langle a,b \rangle$…

$$D_{\vec{v}} f = \nabla f \cdot \frac{\vec{v}}{\|v\|} = \frac{a}{\sqrt{a^2+b^2}} f_x + \frac{b}{\sqrt{a^2+b^2}} f_y$$

Tangent planes…

$$\vec{n} = \langle f_x, f_y, -1 \rangle$$

$$z = f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0) + f(x_0,y_0)$$

Multivariable chain rule: given $z = f(u,v)$: $u(s,t), v(s,t)$…

$$\frac{dz}{dt} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial t} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial t}$$

$$\frac{dz}{ds} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial s} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial s}$$

The point (x_0,y_0) is a critical point on the surface $z = f(x,y)$ if…

$$f_x(x_0,y_0) = 0 \textit{ AND } f_y(x_0,y_0) = 0 \text{ or undefined}$$

Discriminant (second derivative test) of $f(x,y)$ at (x_0,y_0)…

$$D(x_0,y_0) = \left[f_{xx}(x_0,y_0) f_{yy}(x_0,y_0)\right] - \left[f_{xy}(x_0,y_0)\right]^2$$

Second derivative test…

$D(x_0,y_0) > 0$	$f_{xx}(x_0,y_0) > 0$	local min
$D(x_0,y_0) > 0$	$f_{xx}(x_0,y_0) < 0$	local max
$D(x_0,y_0) < 0$		saddle point
$D(x_0,y_0) = 0$		inconclusive

Lagrange multipliers: find extrema of $f(x,y)$ given constraint $g(x,y) = k$…

$$\nabla f = \lambda \nabla g$$

$$= \begin{cases} f_x = \lambda g_x \\ f_y = \lambda g_y \\ g(x,y) = k \end{cases}$$

GIVEN FOR EACH STATEMENT…

In $\mathbb{R}^2$…

$$R = \{(x, y) \mid x_1 \leq x \leq x_2, y_1 \leq y \leq y_2\}$$

$$\mathrm{C} = \{(\mathrm{r}, \theta) \mid \mathrm{r}_1 \leq \mathrm{r} \leq \mathrm{r}_2, \theta_1 \leq \theta \leq \theta_2\}$$

In $\mathbb{R}^3$…

$$R = \{(x, y, z) \mid x_1 \leq x \leq x_2, y_1 \leq y \leq y_2, z_1 \leq z \leq z_2\}$$

$$\mathrm{C} = \{(\mathrm{r}, \theta, \mathrm{z}) \mid \mathrm{r}_1 \leq \mathrm{r} \leq \mathrm{r}_2, \theta_1 \leq \theta \leq \theta_2, z_1 \leq z \leq z_2\}$$

$$S = \{(\rho, \phi, \theta) \mid \rho_1 \leq \rho \leq \rho_2, \phi_1 \leq \phi \leq \phi_2, \theta_1 \leq \theta \leq \theta_2\}: \phi = [-\pi, \pi], \theta = [0, 2\pi]$$

TRANSFORMATIONS BETWEEN COORDINATE SYSTEMS…

General…

$$x^2 + y^2 + z^2 = r^2 \text{ or } x^2 + y^2 + z^2 = \rho^2$$

$$\tan\theta = \frac{y}{x}$$

Cartesian to polar…

$$dx\, dy \rightarrow \begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases} \rightarrow r\, dr\, d\theta$$

Cartesian to cylindrical

$$dz\, dx\, dy \rightarrow \begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases} \rightarrow r\, dz\, dr\, d\theta$$

Cartesian to spherical…

$$dz\, dx\, dy \rightarrow \begin{cases} x = \rho\sin\phi\cos\theta \\ y = \rho\sin\phi\sin\theta \\ r = \rho\sin\phi \end{cases} \rightarrow \rho^2\sin\phi\ d\rho\, d\phi\, d\theta$$

Integrals & applications in different coordinate systems

Double integral of $f(x,y)$ over region R (cartesian)…

$$\iint_R f(x,y)\,dA = \int_c^d \int_a^b f(x,y)\,dx\,dy = \int_a^b \int_c^d f(x,y)\,dy\,dx$$

Average value of $f(x,y)$ over region R…

$$f_{avg} = \frac{1}{area\ of\ R}\iint_R f(x,y)\,dA$$

Area of region R…

$$area\ of\ R = \iint_R 1\,dA$$

Volume of region R with height $f(x,y)$…

$$volume\ of\ R = \iint_R f(x,y)\,dA$$

Double integrals in polar coordinates…

$$\iint_R f(x,y)\,dA = \iint_C f(r,\theta)\cdot r\,dr\,d\theta$$

Volume of region R…

$$volume\ of\ region\ R = \iiint_R 1\,dV$$

Triple integrals in cartesian over R (hypervolume)…

$$\iiint_R f(x,y,z)\,dV = \int_{x_1}^{x_2}\int_{y_1}^{y_2}\int_{z_1}^{z_2} f(x,y,z)\,dz\,dy\,dx = \int_{y_1}^{y_2}\int_{x_1}^{x_2}\int_{z_1}^{z_2} f(x,y,z)\,dz\,dx\,dy$$

Integral in cylindrical coordinates over region C…

$$\iiint_R f(x,y,z)\,dV = \iiint_C f(r,\theta,z)\,r\,dz\,dr\,d\theta = \iiint_C f(r,\theta,z)\,r\,dz\,d\theta\,dr$$

Integral in spherical coordinates over region S…

$$\iiint_R f(x,y,z)\,dV = \iiint_S f(\rho,\phi,\theta)\,(\rho^2\sin\phi)d\rho\,d\phi\,d\theta = \iiint_S f(\rho,\phi,\theta)\,(\rho^2\sin\phi)d\rho\,d\theta\,d\phi$$